

# Abhinav Goel

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*Place of birth:* Muzaffarnagar, Uttar Pradesh, India \* *Date of birth:* 18-12-2004

*LinkedIn:* [Abhinav Goel](#) \* *GitHub:* [Humobot1812](#) \* *Portfolio:* [Humobot1812.github.io/Portfolio](#)

## Education

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### B. Tech. in Electronics and Communication Engineering

PDPM IITDMJ

*Bachelor's Degree Program*

*August 2024 - Present*

*Semester IV (CPI - 8.5)*

### Senior Secondary (Higher Secondary)

PR Public School, Muzaffarnagar

*Central Board of Secondary Education (CBSE)*

*July 2022 - July 2024*

*Percentage: 93.6%*

### Secondary (High School)

Shardein School, Muzaffarnagar

*Central Board of Secondary Education (CBSE)*

*April 2020 - July 2022*

*Percentage: 98.4%*

## Areas of interest

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- Autonomous Mobile Robots (AMRs), SLAM and Navigation (ROS 2, Nav2)
- Unmanned Aerial Vehicles (UAVs) & Autonomous Aerial Flight Systems
- Robotic Manipulation, Visual Servoing & Kinematics
- Computer Vision, Spatial Geolocation & Multi-Spectral Perception
- Tactical Ground Control Stations (GCS) & Human-Machine Interfaces (Qt6 / C++)
- Embedded Systems, Flight Controllers & Discrete Power Electronics
- Subterranean, Field Exploration & Bio-Inspired Legged Robotics

## Honors and key achievements

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- **e-Yantra Robotics Competition (e-YRC 2025–26), IIT Bombay:** Top 10 National Finalist among 400+ participating teams across India in the *KrishiCobot* theme, designing an autonomous agricultural harvesting mobile manipulator with ROS 2, UR5 visual servoing, and precision vision pipelines. [[Certificate](#)]
- **DD Robocon 2025 (Doordarshan & IIT Delhi):** Achieved Stage-2 National Qualification representing Team Barbarick with an evaluation score of 86/100, designing competition-grade holonomic robot chassis and actuation mechanisms. [[Certificate](#)]
- **SAE Aerothon 2025 (SAE India):** Phase 2 National Qualification in the premier aerial robotics challenge, demonstrating an autonomous fixed-altitude surveillance quadcopter with ground target pixel-to-GPS conversion. [[Certificate](#)]
- **NIDAR 2026 (National Innovation Challenge for Drone Application and Research):** National Finalist, Aero Fabrication Club, developing an autonomous disaster surveillance UAV with 75+ acre automated polygon coverage and real-time survivor detection. [[Certificate](#)]

### **Autonomous Agricultural Harvesting Robot (KrishiCobot)**

*August 2025 - May 2026*

*e-Yantra Robotics Competition 2025–26, IIT Bombay*

[github.com/Humobot1812/Krishi\\_Cobot](https://github.com/Humobot1812/Krishi_Cobot)

- Architected and developed a full-stack autonomous harvesting robot integrating an AMR mobile base with a 6-DOF UR5 robotic arm, operating autonomously in complex orchard rows within Gazebo Ignition.
- Constructed a multi-sensor navigation pipeline fusing 2D planar LiDAR, ultrasonic range arrays, IMU, and wheel odometry inside ROS 2 Nav2, utilizing AMCL localization to achieve repeatable navigation in narrow crop corridors.
- Developed a two-tier perception pipeline integrating YOLOv8 for fruit classification and ArUco markers for 6-DOF pose estimation, projecting camera coordinates to 3D robot base frame with millimeter precision.
- Engineered Cartesian trajectory generation via MoveIt 2 alongside Image-Based Visual Servoing (IBVS) to continuously update end-effector paths in real time based on active camera tracking.
- Coordinated subsystems through a ROS 2 action server and behavior tree framework; achieved a Top 10 national rank out of 400+ teams across India.

### **Autonomous Warehouse Stock Auditing AMR & Active Vision System (WareOps)**

*December 2025 - August 2026*

*Robotics & Autonomous Systems Lab, PDPM IITDMJ*

[github.com/WareOps](https://github.com/WareOps)

- Engineered an autonomous mobile robot for automated warehouse stock auditing, featuring autonomous aisle navigation, active mast scanning, and automated digital ledger reporting.
- Configured the ROS 2 Nav2 stack with SLAM Toolbox for initial mapping and AMCL for runtime localization, tuning DWB local planner costmap critics for smooth traversal of narrow rack aisles.
- Designed and fabricated a 2-DOF pan-tilt motorized vision mast controlled by ROS 2 action servers to sweep multiple storage rack tiers during audit waypoints.
- Implemented a dual-engine QR reading pipeline executing OpenCV WeChat QR detector in parallel with PyZbar for high-accuracy barcode and QR decoding on damaged labels, logging results into an SQLite database.
- Maintained 1:1 hardware-simulation parity between Gazebo Ignition and physical hardware comprising Raspberry Pi 4, RPLidar A1M8, Arduino Nano via ros2\_control, and ESP32-CAM Wi-Fi video streaming.

### **Autonomous 6-Wheeled Subterranean Exploration Rover (CAVE\_DIVER)**

*September 2026 - Present*

*Electronics and Robotics Society, PDPM IITDMJ*

[github.com/Humobot1812/CAVE\\_DIVER\\_6\\_Wheeled\\_Rover](https://github.com/Humobot1812/CAVE_DIVER_6_Wheeled_Rover)

- Engineered an autonomous 6-wheeled robotic rover for exploration, 3D structural mapping, and environmental hazard assessment in GPS-denied cave and subterranean environments.
- Designed a rocker-bogie passive suspension mechanism modeled in Autodesk Fusion 360 and 3D-printed in PETG, enabling all 6 driven wheels to maintain continuous terrain contact across obstacles up to 1.5× wheel diameter.
- Developed a multi-spectral perception system featuring a 2-DOF gimbal carrying a low-light Raspberry Pi Camera v3 NoIR alongside a FLIR-compatible thermal sensor for human heat signature and hazard detection.
- Integrated an onboard 6-channel hazardous gas telemetry suite monitoring CO, H<sub>2</sub>S, CH<sub>4</sub>, LPG, and smoke concentrations at 1 Hz with automated threshold alarming and safety interlocks.
- Deployed SLAM Toolbox in lifelong mapping mode for 2D LiDAR-based mapping and pose estimation in GPS-denied subterranean tunnels, paired with a custom web-based Flask Ground Control Station.

## **Tactical Multi-UAV Ground Control Station (DroneGCS)**

*September 2026 - Present*

*Aero Fabrication Club, PDPM IITDMJ*

[github.com/Humobot1812/DroneGCS](https://github.com/Humobot1812/DroneGCS)

- Developed an open-source tactical Ground Control Station in C++17 and Qt 6 / QML for simultaneous command, control, and telemetry tracking of multiple UAVs operating ArduPilot and PX4 autopilots.
- Engineered a real-time Heads-Up Display (HUD) parsing MAVLink v2 telemetry packets (HEARTBEAT, ATTITUDE, GPS\_RAW\_INT, SYS\_STATUS, VFR\_HUD) at 10 Hz, rendering pitch/roll artificial horizon, airspeed, altitude, and heading compass.
- Integrated a dynamic OpenStreetMap interactive tactical map via QtLocation plugin, supporting real-time multi-UAV GPS position tracking, multi-waypoint mission editing, and polygon geofence deployment.
- Implemented asynchronous C++ network backends for MAVLink over UDP/TCP and serial connections, handling automated parameter management (PARAM\_REQUEST\_LIST / PARAM\_SET) and command protocols.
- Integrated GStreamer-accelerated H.264 video pipelines for ultra-low latency live onboard camera streaming with multi-drone telemetry overlay.

## **Beginner-Safe ROS 2 Drone Controller Stack (SafePilot)**

*September 2026 - Present*

*Open-Source UAV Development*

[github.com/Humobot1812/SafePilot](https://github.com/Humobot1812/SafePilot)

- Created an open-source ROS 2 package wrapping MAVSDK to implement software-enforced flight safety envelopes and autonomous flight assistance for research and beginner UAV pilots.
- Developed a joystick teleoperation node with dynamic velocity and attitude clamps that restrict maximum bank angles, descent rates, and horizontal accelerations to prevent vehicle loss-of-control.
- Implemented a 10 Hz 3D cylindrical geofencing monitor that applies progressive resistive velocity damping as the vehicle nears boundary walls and triggers autonomous Return-to-Launch (RTL) upon perimeter breach.
- Integrated autonomous assistance flight modes including altitude-hold with planar joystick nudging and automated landing sequences with precision descent profiling.

## **Autonomous Disaster Surveillance UAV & Victim Detection**

*June 2025 - January 2026*

*NIDAR 2026 National UAV Challenge, Aero Fabrication Club*

[github.com/Humobot1812/Autonomous\\_drone](https://github.com/Humobot1812/Autonomous_drone)

- Developed an autonomous aerial surveillance system for large-area disaster relief and victim localization, capable of autonomously covering 75+ acres within 20 minutes at 25 m altitude.
- Built a computational geometry pipeline using Shapely and KML boundary parsing to generate optimized boustrophedon (zigzag) survey flight paths with parameterized lane overlap.
- Deployed a fine-tuned YOLOv8 human detection model on the companion computer processing downward camera feeds at 15 FPS, triggering autonomous loiter and target-centering routines upon detection.
- Automated the complete mission lifecycle through MAVSDK Python, handling automated pre-arm checks, guided takeoff, survey path traversal, interrupt-driven victim loiter, geofence compliance, and failsafe RTL.
- Conducted extensive Software-In-The-Loop (SITL) validation in ArduPilot SITL with simulated visual streams prior to field trials.

## **Autonomous Aerial Survey & Ground Target Geolocation Drone**

*January 2025 - June 2025*

*SAE Aerothon 2025, Aero Fabrication Club, PDPM IITDMJ*

[github.com/Humobot1812/Aerothon-25](https://github.com/Humobot1812/Aerothon-25)

- Engineered an autonomous UAV platform selected for national-level presentation at SAE Aerothon 2025, integrating a Pixhawk flight controller, Raspberry Pi companion computer, and ArduPilot autopilot.
- Derived and implemented a pinhole camera projection algorithm converting 2D pixel coordinates of detected targets into georeferenced latitude/longitude coordinates accurate to within  $\pm 3$  m from 30 m flight altitude.
- Trained and deployed YOLOv4-Tiny on the companion computer for real-time target panel classification and bounding-box estimation at 12 FPS.
- Developed a visual PID control loop computing lateral and longitudinal velocity commands to center detected targets beneath the vehicle during loiter maneuvers.

## Differential Drive Mobile Robot — ROS 2 Navigation & SLAM

June 2025 - July 2025

Electronics and Robotics Society, PDPM  
IIITDMJ

[github.com/Humobot1812/Differential-Drive-ROS2-Navigation-Project](https://github.com/Humobot1812/Differential-Drive-ROS2-Navigation-Project)

- Modeled and simulated a 4-wheeled differential-drive autonomous robot from scratch using URDF/Xacro, integrating collision models, wheel inertia matrices, LiDAR, and IMU sensor plugins in Gazebo.
- Configured SLAM Toolbox in asynchronous mode to generate 2D occupancy grid maps in simulation mazes, tuning scan-matching thresholds for sharp wall boundaries and minimal odometry drift.
- Implemented AMCL particle filtering for real-time localization on saved maps, tuning covariance parameters for rapid recovery during kidnapping and slippage conditions.
- Configured Nav2 with NavFn global planner and DWB local planner, optimizing obstacle avoidance critics for smooth trajectory generation through narrow doorway clearances.
- Integrated robot\_localization Extended Kalman Filter (EKF) fusing wheel odometry and IMU data to ensure drift-free state estimation; authored a 12-part public tutorial series earning 7 GitHub stars.

## Smart Multi-Protocol ESP32 Robotics Transmitter & Receiver System

April 2026 - July 2026

Electronics and Robotics Society, PDPM  
IIITDMJ

[github.com/Humobot1812/Smart-Multi-Protocol-ESP32-Transmitter](https://github.com/Humobot1812/Smart-Multi-Protocol-ESP32-Transmitter)

- Designed and built a modular dual-ESP32 smart remote control transmitter supporting four switchable wireless protocols: NRF24L01+ (2.4 GHz low latency), ESP-NOW (peer-to-peer 250 m), Wi-Fi UDP, and Bluetooth Classic.
- Developed firmware utilizing the U8g2 graphics library on a 128×64 ST7920 graphical LCD, rendering real-time RSSI signal strength, battery levels, channel bar graphs, and an interactive menu.
- Incorporated an MPU6050 6-axis IMU enabling tilt-based gesture control, mapping physical transmitter pitch and roll directly into motor velocity and steering commands.
- Built a companion modular receiver unit outputting multichannel servo PWM and serial telemetry back to the handheld unit.

## Custom ESP32-S3 Quadcopter Drone with Discrete MOSFET ESCs

January 2026 - February 2026

Aero Fabrication Club, PDPM IIITDMJ

[youtube.com/@The\\_IIITians\\_World](https://youtube.com/@The_IIITians_World)

- Designed and assembled custom 3-phase Electronic Speed Controllers (ESCs) using discrete N-channel MOSFETs and gate drivers, eliminating commercial off-the-shelf ESCs.
- Implemented 6-step trapezoidal commutation firmware on an ESP32-S3 microcontroller, achieving smooth BLDC motor speed control with configurable acceleration ramps.
- Programmed a 500 Hz flight control PID loop reading angular rates and accelerations from an MPU6050 sensor to stabilize roll, pitch, and yaw axes in real time.
- Integrated a BMP280 barometric pressure sensor with accelerometer Z-axis data through a complementary filter for reliable relative altitude estimation and hover stabilization.

## Modular 5-DOF Robotic Arm with Geometric Inverse Kinematics

March 2025 - Present

Electronics and Robotics Society, PDPM IIITDMJ

[github.com/Humobot1812/Robotic\\_arm](https://github.com/Humobot1812/Robotic_arm)

- Designed and 3D-printed a 5-DOF articulated serial robotic arm utilizing high-torque MG996R and DS3225 metal-gear servo actuators.
- Established Denavit-Hartenberg (DH) kinematic frames and derived closed-form geometric inverse kinematics equations for real-time Cartesian point-to-point positioning on an Arduino microcontroller.
- Implemented cubic polynomial trajectory interpolation to ensure smooth acceleration and deceleration profiles, eliminating mechanical vibrations during rapid motion.
- Designed a compliant parallel-jaw gripper capable of securely grasping objects up to 60 mm in diameter.

## Serpentine Snake Robot Prototype with Sinusoidal Locomotion Gaits

June 2025 - July 2025

Electronics and Robotics Society, PDPM IIITDMJ

[github.com/Humobot1812/Snake\\_Robot](https://github.com/Humobot1812/Snake_Robot)

- Designed and constructed an 8-segment bio-inspired snake robot using modular 3D-printed body links interconnected with high-torque servo joints.
- Implemented Hirose serpenoid curve mathematical equations for gait generation, generating traveling sinusoidal waves with controllable amplitude, frequency, and phase offset.
- Evaluated multiple locomotion modes including lateral undulation for planar movement, sidewinding for loose terrain, and concertina locomotion for crawling through confined PVC pipes.

- Engineered an automated student check-in/check-out system deployed in the institute hostel, utilizing MFRC522 RFID reader hardware interfaced with ESP8266 Wi-Fi microcontrollers.
- Developed a Python Flask REST backend backed by SQLite to record real-time gate entry and exit events with cryptographic timestamp verification.
- Built a responsive administrative dashboard providing hostel authorities with real-time student presence counts, entry violation alerts, and downloadable attendance reports.

Certifications

- e-Yantra Robotics Competition (e-YRC 2025–26) Finalist:** Awarded by IIT Bombay for securing Top 10 nationally among 400+ collegiate teams in the autonomous KrishiCobot agricultural robotics theme (August 2025 – May 2026).
- DD Robocon 2025 Stage-2 Certificate:** Awarded by Doordarshan & IIT Delhi for qualifying Stage 2 with Team Barbarick, scoring 86/100 in technical design and mechanism evaluations (2025).
- SAE Aerothon 2025 National Aerial Robotics Certificate:** Awarded by SAE India for qualifying Phase 2 in the national autonomous UAV surveillance challenge (June 2025).
- NIDAR 2026 National UAV Challenge:** Certified participant in National Innovation Challenge for Drone Application and Research (January 2026).
- Hands-on Introduction to Autonomous Systems with ROS 2 & Embedded Hardware:** PDPM IITDMJ Student Technical Workshop (2024).

Technical Skills

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|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Robotics & Autonomous Systems:  | ROS 2 (Humble), Nav2, ros2_control, SLAM Toolbox, AMCL, TF2, URDF/Xacro, MoveIt 2, Kinematics, Trajectory Planning, Sensor Fusion, EKF, RANSAC          |
| UAV & Flight Systems:           | ArduPilot, PX4, Pixhawk, MAVLink, MAVSDK, DroneKit, QGroundControl, Mission Planner, ArduPilot SITL, PID Control                                        |
| Computer Vision & Perception:   | OpenCV, YOLOv8, YOLOv4-Tiny, ArUco, Visual Servoing, Pinhole Camera Model, ArUco Marker Geolocation, ORB-SLAM3                                          |
| Embedded Systems & Electronics: | ESP32-S3, ESP32-CAM, Arduino, STM32, UART, I2C, SPI, PWM, MPU9250/MPU6050, Encoder Motors, Barometer, NRF24L01, LoRa, ST7920 Display, Custom MOSFET ESC |
| Compute, Sensors & Hardware:    | Raspberry Pi 4/5, NVIDIA Jetson Nano/AGX Xavier, RPLidar A1M8, LiDAR, Raspberry Pi Camera v3 NoIR                                                       |
| Simulation & Engineering Tools: | Gazebo, RViz2, MuJoCo, MATLAB/Simulink, Fusion 360, 3D Printing, TinkerCAD, Git, Linux (Ubuntu 22.04)                                                   |
| Programming Languages:          | Python, C++                                                                                                                                             |

Language proficiency

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|----------------|-------------------------------|
| English, Hindi | Full Professional Proficiency |
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Memberships

- Electronics and Robotics Society of PDPM IITDMJ** (Coordinator, March 2026 – Present | Member, August 2024 – March 2026)
- Aero Fabrication Club of PDPM IITDMJ** (Core Committee Member, May 2025 – Present | Member, August 2024 – May 2025)